Simple Identities Pt. 1

1. $E[f(W) + g(U)] = E[f(W)] + E[g(U)]$ for any two random variables U and W

Here W and U are the random variables, w and u are their possible values, and f and g are the given functions.

Because the expression depends on both variables, use their joint probability density function (pdf), $p_{WU}(w, u)$, as defined for two variables in *Topic 2.pdf*, slide 30. This density describes the values of W and U together. Integrating the pdf over a range gives the probability of values in that range.

Using the joint expectation formula in *Topic 2.pdf*, slide 40, gives

$$E[f(W) + g(U)] = \int_{-\infty}^{\infty} dw\, du\, [f(w) + g(u)] p_{WU}(w, u). \tag{1}$$

Multiplying the density, $p_{WU}(w, u)$, into both terms gives

$$[f(w) + g(u)] p_{WU}(w, u) = f(w) p_{WU}(w, u) + g(u) p_{WU}(w, u).$$

We can therefore split the integral in equation (1):

$$E[f(W) + g(U)] = \int_{-\infty}^{\infty} dw\, du\, f(w) p_{WU}(w, u) + \int_{-\infty}^{\infty} dw\, du\, g(u) p_{WU}(w, u). \tag{2}$$

For the first term of equation (2), integrate over u while keeping w fixed. Because w is fixed, $f(w)$ is a constant during this integration:

$$\int_{-\infty}^{\infty} f(w) p_{WU}(w, u)\, du = f(w) \int_{-\infty}^{\infty} p_{WU}(w, u)\, du.$$

Here $p_W(w)$ means the pdf of W alone. The marginal pdf formula in *Topic 2.pdf*, slide 33 says that integrating the joint pdf over all values of u gives this pdf:

$$\int_{-\infty}^{\infty} p_{WU}(w, u)\, du = p_W(w). \tag{3}$$

Use equation (3) to replace the integral on the right:

$$f(w) \int_{-\infty}^{\infty} p_{WU}(w, u)\, du = f(w) p_W(w).$$

Thus the result of the integration over u is

$$\int_{-\infty}^{\infty} f(w) p_{WU}(w, u)\, du = f(w) p_W(w). \tag{4}$$

Equation (4) finishes the integration over u. The first term of equation (2) also includes an integration over w, which we still have to do. Replace its integration over u by the result $f(w)p_W(w)$ from equation (4):

$$\int_{-\infty}^{\infty} dw\, du\, f(w) p_{WU}(w, u) = \int_{-\infty}^{\infty} f(w) p_W(w)\, dw. \tag{5}$$

Now, we remember the expectation definition from *Topic 2.pdf*, slide 24:

$$E[f(W)] = \int_{-\infty}^{\infty} f(w) p_W(w)\, dw. \tag{6}$$

Equation (6) gives the integral on the right of equation (5) as $E[f(W)]$. Making this replacement gives the first term to be:

$$\int_{-\infty}^{\infty} dw\, du\, f(w) p_{WU}(w,u) = E[f(W)]. \tag{7}$$

Now, for the second term of equation (2), integrate over w while keeping u fixed. Following the same procedure used for the first term, take $g(u)$ outside the integral over w, use the marginal pdf formula to obtain $p_U(u)$, and then apply the expectation definition. This gives

$$\int_{-\infty}^{\infty} dw\, du\, g(u) p_{WU}(w,u) = E[g(U)]. \tag{8}$$

Return to equation (2). Replace the results from (7) and (8) to get

$$\boxed{E[f(W) + g(U)] = E[f(W)] + E[g(U)]} \tag{9}$$

2. $E[X + a] = E[X] + a$, where a is a constant while X is a random variable.

Here we let a be a constant and X a random variable with pdf $p_X(x)$, where x is a possible value of X. The expectation definition is given as

$$E[f(X)] = \int_{-\infty}^{\infty} f(x)p_X(x)\,dx. \tag{10}$$

If we choose $f(x) = x + a$ in equation (10) gives

$$E[X + a] = \int_{-\infty}^{\infty} (x + a)p_X(x)\,dx.$$

Multiply $p_X(x)$ into the two terms inside the parentheses:

$$(x + a)p_X(x) = xp_X(x) + ap_X(x).$$

Put this expression into the integral:

$$E[X + a] = \int_{-\infty}^{\infty} [xp_X(x) + ap_X(x)]\,dx.$$

Split the integral of the sum into a sum of integrals:

$$E[X + a] = \int_{-\infty}^{\infty} xp_X(x)\,dx + \int_{-\infty}^{\infty} ap_X(x)\,dx.$$

For the first integral, use the expectation definition from *Topic 2.pdf*, slide 24 with $f(x) = x$. Here X is the random variable, while x represents a possible numerical value of X. The expectation definition is

$$E[f(X)] = \int_{-\infty}^{\infty} f(x)p_X(x)\,dx.$$

Choose $f(x) = x$, meaning that the function returns its input. With the random variable as its input, this same function gives $f(X) = X$.

Replace $f(X)$ by X on the left:

$$E[X] = \int_{-\infty}^{\infty} f(x)p_X(x)\,dx.$$

Then replace $f(x)$ by x on the right:

$$E[X] = \int_{-\infty}^{\infty} xp_X(x)\,dx.$$

Replace only the first integral using this definition:

$$E[X + a] = E[X] + \int_{-\infty}^{\infty} ap_X(x)\,dx.$$

Since a does not depend on x, take it outside the remaining integral:

$$E[X + a] = E[X] + a\int_{-\infty}^{\infty} p_X(x)\,dx.$$

The total area formula for a pdf in *Topic 2.pdf*, slide 13 is

$$\int_{-\infty}^{\infty} p_X(x)\,dx = 1. \tag{11}$$

Use equation (11) to replace the remaining integral by 1:

$$E[X + a] = E[X] + a \cdot 1.$$

Multiplying a by 1 leaves a, so

$$\boxed{E[X + a] = E[X] + a.} \tag{12}$$

3. $E[aX] = aE[X]$

Here we let a be a constant and X a random variable with pdf $p_X(x)$, where x is a possible value of X. Remembering that the expectation definition is

$$E[f(X)] = \int_{-\infty}^{\infty} f(x)p_X(x)\,dx.$$

Choose $f(x) = ax$. Substituting this choice gives

$$E[aX] = \int_{-\infty}^{\infty} axp_X(x)\,dx.$$

Writing the integrand as $a[xp_X(x)]$ to show that a multiplies the whole expression being integrated:

$$E[aX] = \int_{-\infty}^{\infty} a[xp_X(x)]\,dx.$$

Since a does not depend on x, take it outside the integral:

$$E[aX] = a\int_{-\infty}^{\infty} xp_X(x)\,dx.$$

Now, we use the expectation definition from *Topic 2.pdf*, slide 24:

$$E[f(X)] = \int_{-\infty}^{\infty} f(x)p_X(x)\,dx.$$

Choose $f(x) = x$. This also means $f(X) = X$, so the definition becomes

$$E[X] = \int_{-\infty}^{\infty} xp_X(x)\,dx.$$

Replace the integral by $E[X]$ to get:

$$\boxed{E[aX] = aE[X].} \tag{13}$$

4. $\text{var}(aX) = a^2\,\text{var}[X]$**, where var[] means variance.**

Here a is a constant and X is a random variable. The variance definition is given as a hint in the homework slides as:

$$\text{var}(X) = E[(X - E[X])^2]. \tag{14}$$

Use this definition with aX instead of X:

$$\text{var}(aX) = E[(aX - E[aX])^2].$$

We remember the result we proved in equation (13), is

$$E[aX] = aE[X].$$

So replacing $E[aX]$ using this result:

$$\text{var}(aX) = E[(aX - aE[X])^2].$$

Both terms inside the parentheses contain the factor a, so we can factor it out like this:

$$aX - aE[X] = a(X - E[X]).$$

Make this replacement inside the variance calculation:

$$\text{var}(aX) = E\left[\big(a(X - E[X])\big)^2\right].$$

The square applies to both a and $(X - E[X])$, so

$$\big(a(X - E[X])\big)^2 = a^2(X - E[X])^2.$$

Substitute this squared expression:

$$\text{var}(aX) = E[a^2(X - E[X])^2].$$

Now, equation (13) applies to any constant multiplying any random variable. Here, a^2 is constant and the expression $(X - E[X])^2$ is a random variable because its value depends on X.

Therefore, applying the same rule gives

$$E[a^2(X - E[X])^2] = a^2E[(X - E[X])^2].$$

Using this equality in our expression for $\text{var}(aX)$, gets us:

$$\text{var}(aX) = a^2E[(X - E[X])^2].$$

The entire expression $E[(X - E[X])^2]$ equals $\text{var}(X)$, as seen here

$$\text{var}(aX) = a^2 \underbrace{E[(X - E[X])^2]}_{\text{var}(X)}.$$

So, now we replace the marked expression by $\text{var}(X)$, keeping the factor a^2 outside:

$$\boxed{\text{var}(aX) = a^2\,\text{var}(X).}$$

5. $\text{var}(Z) = \text{var}(X) + \text{var}(Y) + 2C_{XY}$**, where** $Z = X + Y$

Let X and Y be random variables and define their sum by

$$Z = X + Y.$$

First we find the expectation of Z. Replacing Z by its definition gives

$$E[Z] = E[X + Y].$$

Equation (9) proved that the expectation of a sum is the sum of the expectations. With the functions chosen to be the variables themselves, that result is

$$E[X + Y] = E[X] + E[Y].$$

So we can use this result to obtain

$$E[Z] = E[X] + E[Y]. \tag{15}$$

Also remembering the variance definition mentioned in gives

$$\text{var}(Z) = E[(Z - E[Z])^2].$$

Replacing Z inside the square by $X + Y$:

$$\text{var}(Z) = E[((X + Y) - E[Z])^2].$$

Now replace $E[Z]$ using equation (15):

$$\text{var}(Z) = E[((X + Y) - (E[X] + E[Y]))^2].$$

The minus sign in front of the second parentheses changes the signs of both terms inside them:

$$\text{var}(Z) = E[(X + Y - E[X] - E[Y])^2].$$

Reordering the terms:

$$\text{var}(Z) = E[(X - E[X] + Y - E[Y])^2].$$

Grouping the two differences gives

$$\text{var}(Z) = E[((X - E[X]) + (Y - E[Y]))^2].$$

To simplify the writing, we define

$$A = X - E[X], \qquad B = Y - E[Y].$$

Using these definitions, we can write

$$\text{var}(Z) = E[(A + B)^2].$$

First expand the square:

$$(A + B)^2 = A^2 + 2AB + B^2.$$

Substitute this expansion into the expression for $\text{var}(Z)$:

$$\text{var}(Z) = E[A^2 + 2AB + B^2].$$

The identity in equation (9) allows us to separate the expectation of a sum.

$$\text{var}(Z) = E[A^2] + E[2AB] + E[B^2].$$

The identity in equation (13) allows us to take the constant 2 outside the expectation:

$$\text{var}(Z) = E[A^2] + 2E[AB] + E[B^2]. \tag{16}$$

We can now identify these three expectations. The variance definition is

$$\text{var}(X) = E[(X - E[X])^2].$$

Since $A = X - E[X]$, the expression inside this expectation is A^2. Therefore,

$$E[A^2] = \text{var}(X). \tag{17}$$

The same variance definition applied to Y gives

$$\text{var}(Y) = E[(Y - E[Y])^2].$$

Since $B = Y - E[Y]$, this gives

$$E[B^2] = \text{var}(Y). \tag{18}$$

The covariance definition in *Lab Topic 2_1.pdf*, slide 5 is

$$C_{XY} = E[(X - E[X])(Y - E[Y])]. \tag{19}$$

Using the definitions of A and B, the product inside this expectation is AB. Thus,

$$C_{XY} = E[AB]. \tag{20}$$

Now, returning to equation (16). First, use equation (17) to replace $E[A^2]$ by $\text{var}(X)$:

$$\text{var}(Z) = \text{var}(X) + 2E[AB] + E[B^2].$$

Next, use equation (18) to replace $E[B^2]$ by $\text{var}(Y)$:

$$\text{var}(Z) = \text{var}(X) + 2E[AB] + \text{var}(Y).$$

Finally, use equation (20) to replace $E[AB]$ by C_{XY}:

$$\text{var}(Z) = \text{var}(X) + 2C_{XY} + \text{var}(Y).$$

Reorder the terms:

$$\boxed{\text{var}(Z) = \text{var}(X) + \text{var}(Y) + 2C_{XY}.} \tag{21}$$

Since $Z = X + Y$, this is also the formula for $\text{var}(X + Y)$.

6. Show that the PDF of the sum of two independent Gaussian random variables is also Gaussian.

We want to find the pdf of $Z = X + Y$ and show that it has the Gaussian form.

Let X and Y be independent Gaussian random variables with means μ_X and μ_Y and positive standard deviations σ_X and σ_Y. The lowercase letters x, y, and z are possible values of X, Y, and Z.

Write f_X and f_Y for their pdfs, and f_{X+Y} for the pdf of their sum. Our starting point is the convolution formula given in *Lab Topic 2_1.pdf*, slide 3:

$$f_{X+Y}(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x)\, dx. \tag{22}$$

For a fixed sum z, if $X = x$, then $Y = z - x$. This explains the argument $z - x$ in the second pdf. We will now insert the two Gaussian pdfs into this formula and show that the result has the same Gaussian form.

Using the Gaussian pdf from *Topic 2.pdf*, slide 21, we write the pdfs of X and Y with their respective means and standard deviations:

$$\begin{aligned} f_X(x) &= \frac{1}{\sqrt{2\pi\sigma_X^2}} \exp\left[-\frac{(x-\mu_X)^2}{2\sigma_X^2}\right], \\ f_Y(y) &= \frac{1}{\sqrt{2\pi\sigma_Y^2}} \exp\left[-\frac{(y-\mu_Y)^2}{2\sigma_Y^2}\right]. \end{aligned} \tag{23}$$

In equation (22), the second pdf is evaluated at $z - x$. Replace y by $z - x$ in equation (23):

$$f_Y(z-x) = \frac{1}{\sqrt{2\pi\sigma_Y^2}} \exp\left[-\frac{(z-x-\mu_Y)^2}{2\sigma_Y^2}\right].$$

Insert $f_X(x)$ and $f_Y(z-x)$ into equation (22). The constant factors do not depend on x, so we can take them outside the integral:

$$f_{X+Y}(z) = \frac{1}{2\pi\sigma_X\sigma_Y} \int_{-\infty}^{\infty} \exp\left[-\frac{(x-\mu_X)^2}{2\sigma_X^2}\right] \exp\left[-\frac{(z-x-\mu_Y)^2}{2\sigma_Y^2}\right] dx.$$

Multiplying exponentials adds their exponents, like this

$$f_{X+Y}(z) = \frac{1}{2\pi\sigma_X\sigma_Y} \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2}\left(\frac{(x-\mu_X)^2}{\sigma_X^2} + \frac{(z-x-\mu_Y)^2}{\sigma_Y^2}\right)\right] dx. \tag{24}$$

Use u as a shorter name for x minus its mean:

$$u = x - \mu_X.$$

Adding μ_X to both sides gives

$$x = u + \mu_X.$$

The first squared difference is therefore

$$(x-\mu_X)^2 = u^2.$$

For the second difference, replace x by $u + \mu_X$:

$$z - x - \mu_Y = z - (u + \mu_X) - \mu_Y.$$

Remove the parentheses:

$$z - x - \mu_Y = z - u - \mu_X - \mu_Y.$$

Move the term $-u$ to the end:

$$z - x - \mu_Y = (z - \mu_X - \mu_Y) - u.$$

Give the expression in parentheses the shorter name r:

$$r = z - \mu_X - \mu_Y.$$

For a chosen value of z, r is a fixed number. Replacing the parentheses by r gives

$$z - x - \mu_Y = r - u.$$

Squaring both sides gives

$$(z - x - \mu_Y)^2 = (r - u)^2.$$

The dx at the end of the original integral tells us to integrate over x. We are now using u, so we write du instead. Now, we put the two squared differences into equation (24), and use du:

$$f_{X+Y}(z) = \frac{1}{2\pi\sigma_X\sigma_Y} \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2}\left(\frac{u^2}{\sigma_X^2} + \frac{(r-u)^2}{\sigma_Y^2}\right)\right] du. \tag{25}$$

If the two means are zero, then $u = x$ and $r = z$. We now rewrite the expression inside the exponent as a square.

$$(r - u)^2 = r^2 - 2ru + u^2.$$

So, now we can say:

$$\frac{u^2}{\sigma_X^2} + \frac{(r-u)^2}{\sigma_Y^2} = \frac{u^2}{\sigma_X^2} + \frac{r^2 - 2ru + u^2}{\sigma_Y^2}.$$

$$\frac{u^2}{\sigma_X^2} + \frac{(r-u)^2}{\sigma_Y^2} = \frac{u^2}{\sigma_X^2} + \frac{r^2}{\sigma_Y^2} - \frac{2ru}{\sigma_Y^2} + \frac{u^2}{\sigma_Y^2}.$$

We will combine the two terms containing u^2. Give them the common denominator $\sigma_X^2\sigma_Y^2$ by multiplying the numerator and denominator by the same quantity:

$$\frac{u^2}{\sigma_X^2} = \frac{\sigma_Y^2 u^2}{\sigma_X^2\sigma_Y^2}.$$

$$\frac{u^2}{\sigma_Y^2} = \frac{\sigma_X^2 u^2}{\sigma_X^2\sigma_Y^2}.$$

Now add them:

$$\frac{u^2}{\sigma_X^2} + \frac{u^2}{\sigma_Y^2} = \frac{\sigma_Y^2 u^2 + \sigma_X^2 u^2}{\sigma_X^2\sigma_Y^2}.$$

Factor u^2 out of the numerator:

$$\frac{u^2}{\sigma_X^2} + \frac{u^2}{\sigma_Y^2} = \frac{(\sigma_X^2 + \sigma_Y^2)u^2}{\sigma_X^2\sigma_Y^2}.$$

The sum of the two variances now appears in this expression. Give that sum the shorter name S:

$$S = \sigma_X^2 + \sigma_Y^2. \tag{26}$$

Both variances are positive, so $S > 0$. Using this name gives

$$\frac{u^2}{\sigma_X^2} + \frac{u^2}{\sigma_Y^2} = \frac{Su^2}{\sigma_X^2\sigma_Y^2}.$$

Now, we put the combined u^2 term back into the full expression:

$$\frac{u^2}{\sigma_X^2} + \frac{(r-u)^2}{\sigma_Y^2} = \frac{Su^2}{\sigma_X^2\sigma_Y^2} - \frac{2ru}{\sigma_Y^2} + \frac{r^2}{\sigma_Y^2}. \tag{27}$$

Now, we let m be a constant whose value we will choose so that $\frac{S(u-m)^2}{\sigma_X^2\sigma_Y^2}$ has the same u^2 and u terms as equation (27). First, we expand:

$$(u-m)^2 = u^2 - 2mu + m^2.$$

Multiply the expansion by its factor:

$$\frac{S(u-m)^2}{\sigma_X^2\sigma_Y^2} = \frac{S(u^2 - 2mu + m^2)}{\sigma_X^2\sigma_Y^2}.$$

Distribute that factor to each term:

$$\frac{S(u-m)^2}{\sigma_X^2\sigma_Y^2} = \frac{Su^2}{\sigma_X^2\sigma_Y^2} - \frac{2Smu}{\sigma_X^2\sigma_Y^2} + \frac{Sm^2}{\sigma_X^2\sigma_Y^2}.$$

We compare the expanded square with equation (27). Both contain the same term in u^2, $\frac{Su^2}{\sigma_X^2\sigma_Y^2}$. The term containing u to the first power in equation (27) is

$$-\frac{2ru}{\sigma_Y^2}.$$

The corresponding term in the expanded square is

$$-\frac{2Smu}{\sigma_X^2\sigma_Y^2}.$$

We choose m so that these two terms are equal for every u. To do this, set the factors multiplying u equal:

$$-\frac{2Sm}{\sigma_X^2\sigma_Y^2} = -\frac{2r}{\sigma_Y^2}.$$

Multiply both sides by $\sigma_X^2\sigma_Y^2$. The denominators cancel, giving

$$-2Sm = -2r\sigma_X^2.$$

Divide both sides by -2:

$$Sm = \sigma_X^2 r.$$

Since $S > 0$, divide both sides by S:

$$m = \frac{\sigma_X^2 r}{S}. \tag{28}$$

With this choice of m, the expanded square has the same u^2 and u terms as equation (27). We still need to account for the terms that do not contain u.

With this choice, the expansion of the square is

$$\frac{S(u-m)^2}{\sigma_X^2\sigma_Y^2} = \frac{Su^2}{\sigma_X^2\sigma_Y^2} - \frac{2ru}{\sigma_Y^2} + \frac{Sm^2}{\sigma_X^2\sigma_Y^2}.$$

Subtract the last term from both sides:

$$\frac{S(u-m)^2}{\sigma_X^2\sigma_Y^2} - \frac{Sm^2}{\sigma_X^2\sigma_Y^2} = \frac{Su^2}{\sigma_X^2\sigma_Y^2} - \frac{2ru}{\sigma_Y^2}.$$

The right side is the first two terms of equation (27). Replace those two terms and keep the remaining term:

$$\frac{u^2}{\sigma_X^2} + \frac{(r-u)^2}{\sigma_Y^2} = \frac{S(u-m)^2}{\sigma_X^2\sigma_Y^2} - \frac{Sm^2}{\sigma_X^2\sigma_Y^2} + \frac{r^2}{\sigma_Y^2}.$$

We still need to simplify the two terms outside the square. First calculate the term containing m^2. Substitute m from equation (28):

$$\frac{Sm^2}{\sigma_X^2\sigma_Y^2} = \frac{S}{\sigma_X^2\sigma_Y^2}\left(\frac{\sigma_X^2 r}{S}\right)^2.$$

Square the numerator and denominator:

$$\frac{Sm^2}{\sigma_X^2\sigma_Y^2} = \frac{S}{\sigma_X^2\sigma_Y^2}\frac{\sigma_X^4 r^2}{S^2}.$$

Multiply the fractions:

$$\frac{Sm^2}{\sigma_X^2\sigma_Y^2} = \frac{S\sigma_X^4 r^2}{\sigma_X^2\sigma_Y^2 S^2}.$$

Cancel one factor of S from the numerator and denominator:

$$\frac{Sm^2}{\sigma_X^2\sigma_Y^2} = \frac{\sigma_X^4 r^2}{\sigma_X^2\sigma_Y^2 S}.$$

Cancel one factor of σ_X^2:

$$\frac{Sm^2}{\sigma_X^2\sigma_Y^2} = \frac{\sigma_X^2 r^2}{S\sigma_Y^2}.$$

Use this result in the two terms outside the square, writing the positive term first:

$$\frac{r^2}{\sigma_Y^2} - \frac{Sm^2}{\sigma_X^2\sigma_Y^2} = \frac{r^2}{\sigma_Y^2} - \frac{\sigma_X^2 r^2}{S\sigma_Y^2}.$$

Give the first fraction the same denominator as the second:

$$\frac{r^2}{\sigma_Y^2} - \frac{Sm^2}{\sigma_X^2\sigma_Y^2} = \frac{Sr^2}{S\sigma_Y^2} - \frac{\sigma_X^2 r^2}{S\sigma_Y^2}.$$

Subtract the numerators:

$$\frac{r^2}{\sigma_Y^2} - \frac{Sm^2}{\sigma_X^2\sigma_Y^2} = \frac{Sr^2 - \sigma_X^2 r^2}{S\sigma_Y^2}.$$

Factor r^2 out of the numerator:

$$\frac{r^2}{\sigma_Y^2} - \frac{Sm^2}{\sigma_X^2\sigma_Y^2} = \frac{r^2(S - \sigma_X^2)}{S\sigma_Y^2}.$$

To simplify the difference in parentheses, use the definition of S in equation (26):

$$S - \sigma_X^2 = (\sigma_X^2 + \sigma_Y^2) - \sigma_X^2.$$

The two σ_X^2 terms cancel:

$$S - \sigma_X^2 = \sigma_Y^2.$$

Replace that difference in the numerator:

$$\frac{r^2}{\sigma_Y^2} - \frac{Sm^2}{\sigma_X^2\sigma_Y^2} = \frac{r^2\sigma_Y^2}{S\sigma_Y^2}.$$

Cancel σ_Y^2:

$$\frac{r^2}{\sigma_Y^2} - \frac{Sm^2}{\sigma_X^2\sigma_Y^2} = \frac{r^2}{S}.$$

Put this result beside the square to obtain

$$\frac{u^2}{\sigma_X^2} + \frac{(r-u)^2}{\sigma_Y^2} = \frac{S(u-m)^2}{\sigma_X^2\sigma_Y^2} + \frac{r^2}{S}. \tag{29}$$

Use equation (29) in equation (25). The exponential containing only r is constant with respect to u, so take it outside the integral:

$$f_{X+Y}(z) = \frac{1}{2\pi\sigma_X\sigma_Y} \exp\left[-\frac{r^2}{2S}\right] \int_{-\infty}^{\infty} \exp\left[-\frac{S(u-m)^2}{2\sigma_X^2\sigma_Y^2}\right] du. \tag{30}$$

The remaining exponential has the Gaussian form in equation (23), with mean m and variance $\frac{\sigma_X^2\sigma_Y^2}{S}$. The Gaussian pdf formula includes the factor

$$\frac{1}{\sqrt{\frac{2\pi\sigma_X^2\sigma_Y^2}{S}}}.$$

Multiplying the exponential by this factor gives a complete Gaussian pdf. Since a pdf has total area 1, its integral is

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{\frac{2\pi\sigma_X^2\sigma_Y^2}{S}}} \exp\left[-\frac{S(u-m)^2}{2\sigma_X^2\sigma_Y^2}\right] du = 1.$$

Multiply by that constant denominator to find the integral we need:

$$\int_{-\infty}^{\infty} \exp\left[-\frac{S(u-m)^2}{2\sigma_X^2\sigma_Y^2}\right] du = \sqrt{\frac{2\pi\sigma_X^2\sigma_Y^2}{S}}. \tag{31}$$

Substitute equation (31) into equation (30):

$$f_{X+Y}(z) = \frac{1}{2\pi\sigma_X\sigma_Y}\exp\left[-\frac{r^2}{2S}\right]\sqrt{\frac{2\pi\sigma_X^2\sigma_Y^2}{S}}.$$

Since σ_X and σ_Y are positive, the constant simplifies to

$$\frac{1}{2\pi\sigma_X\sigma_Y}\sqrt{\frac{2\pi\sigma_X^2\sigma_Y^2}{S}} = \frac{\sqrt{2\pi}\sigma_X\sigma_Y}{2\pi\sigma_X\sigma_Y\sqrt{S}} = \frac{1}{\sqrt{2\pi S}}.$$

Thus

$$f_{X+Y}(z) = \frac{1}{\sqrt{2\pi S}}\exp\left[-\frac{r^2}{2S}\right].$$

Finally, put back $r = z - (\mu_X + \mu_Y)$ and $S = \sigma_X^2 + \sigma_Y^2$:

$$\boxed{f_{X+Y}(z) = \frac{1}{\sqrt{2\pi(\sigma_X^2+\sigma_Y^2)}}\exp\left[-\frac{(z-(\mu_X+\mu_Y))^2}{2(\sigma_X^2+\sigma_Y^2)}\right].} \tag{32}$$

Compare this with the Gaussian pdf in equation (23). The mean is the value subtracted from z, $\mu_X+\mu_Y$. The variance is $\sigma_X^2 + \sigma_Y^2$. Therefore, the sum of the two independent Gaussian variables is Gaussian.

Source: MIT OpenCourseWare: 12.4 The Sum of Independent Normal Random Variables.

7. Covariance of two statistically independent random variables is always zero

Following the hint, first we need to show that $E[X - E[X]] = 0$. Since $E[X]$ is a constant, we can use the result $E[X + a] = E[X] + a$ from equation (12) with $a = -E[X]$:

$$E[X - E[X]] = E[X] + (-E[X]) = E[X] - E[X] = 0. \tag{33}$$

The same argument for Y gives

$$E[Y - E[Y]] = E[Y] - E[Y] = 0. \tag{34}$$

Now let X and Y be independent. The covariance definition already written in equation (19), is

$$C_{XY} = E[(X - E[X])(Y - E[Y])].$$

Let $p_{XY}(x, y)$ be the joint pdf and $p_X(x)$ and $p_Y(y)$ the individual pdfs. Here x and y are possible values of X and Y. As per *Topic 2.pdf*, slide 38, random variables X and Y are said to be statistically independent if:

$$p_{XY}(x, y) = p_X(x)p_Y(y). \tag{35}$$

We start with the covariance definition and to write this expectation as an integral, we use the formula from *Topic 2.pdf*, slide 40. For a function f of two random variables, the formula is

$$E[f(X, Y)] = \int_{-\infty}^{\infty} dx\, dy\, f(x, y)p_{XY}(x, y).$$

Here $f(X, Y)$ is the expression we want to average. The lowercase letters x and y represent possible values of X and Y. The joint pdf $p_{XY}(x, y)$ provides the density used to calculate that average.

For covariance, the expression being averaged is

$$f(X, Y) = (X - E[X])(Y - E[Y]).$$

At the possible values x and y, this expression becomes

$$f(x, y) = (x - E[X])(y - E[Y]).$$

The means $E[X]$ and $E[Y]$ stay the same because they are fixed numbers.

Substitute this expression into the expectation formula:

$$E[(X - E[X])(Y - E[Y])] = \int_{-\infty}^{\infty} dx\, dy\, (x - E[X])(y - E[Y])p_{XY}(x, y).$$

The expectation on the left is the covariance, so

$$C_{XY} = \int_{-\infty}^{\infty} dx\, dy\, (x - E[X])(y - E[Y])p_{XY}(x, y).$$

Because X and Y are independent, *Topic 2.pdf*, slide 38 gives

$$p_{XY}(x, y) = p_X(x)p_Y(y).$$

Replace $p_{XY}(x, y)$ by this product:

$$C_{XY} = \int_{-\infty}^{\infty} dx\, dy\, (x - E[X])(y - E[Y])p_X(x)p_Y(y).$$

Using equation (35) to replace the joint pdf, then group the factors by their variables:

$$C_{XY} = \int_{-\infty}^{\infty} dx\, dy\, [(x - E[X])p_X(x)]\, [(y - E[Y])p_Y(y)]. \tag{36}$$

Integrate over y first. The factor $(x - E[X])p_X(x)$ is constant with respect to y, so it stays outside that integration. The expectation definition in *Topic 2.pdf*, slide 24 gives

$$\int_{-\infty}^{\infty} (y - E[Y])p_Y(y)\, dy = E[Y - E[Y]].$$

Use this result in equation (36). The remaining integration is over x:

$$C_{XY} = \int_{-\infty}^{\infty} (x - E[X])p_X(x)E[Y - E[Y]]\, dx.$$

The number $E[Y - E[Y]]$ is constant with respect to x. Take it outside:

$$C_{XY} = E[Y - E[Y]] \int_{-\infty}^{\infty} (x - E[X])p_X(x)\, dx.$$

The same expectation definition from *Topic 2.pdf*, slide 24 identifies the remaining integral:

$$\int_{-\infty}^{\infty} (x - E[X])p_X(x)\, dx = E[X - E[X]].$$

Thus,

$$C_{XY} = E[Y - E[Y]]E[X - E[X]].$$

Both factors are zero by equations (34) and (33). Substituting those values proves

$$\boxed{C_{XY} = 0 \cdot 0 = 0.} \tag{37}$$

Source: The Book of Statistical Proofs: Covariance of independent random variables.